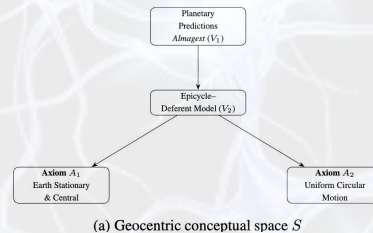


TwistBench : Benchmarking Transformational Creativity in LLMs via Literary Plot Twists



Samuel Schapiro*
AI Innovation Institute
Knowledge Lab



Alexi Gladstone
UIUC



Heng Ji
UIUC



Roger E. Beaty
Penn State

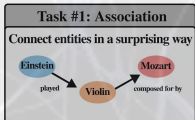
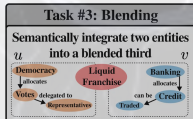
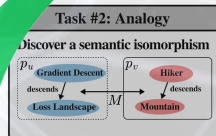
Types of Creative Processes & Products

Boden's types



Combinatorial Creativity

Generate novel ideas that are unfamiliar combinations of familiar ideas



Exploratory Creativity

Generate novel ideas by following existing rules



Transformational Creativity

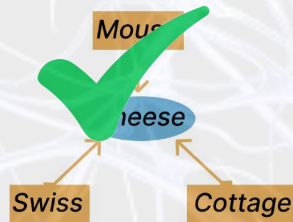
Re-define the rules to produce new possibilities and interpretations of existing ideas (gestalt switch)



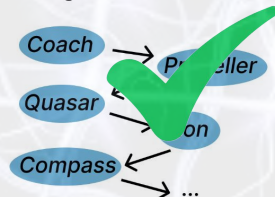
Divergent vs. convergent thinking



Remote Associates Test



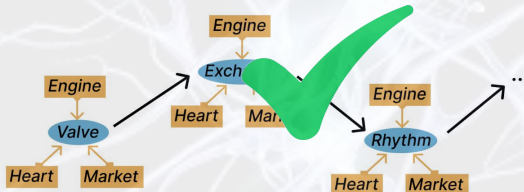
Divergent Association Task



Divergent thinking

Convergent thinking

★ Divergent Remote Association Test



Convergent + Divergent thinking

Abduction

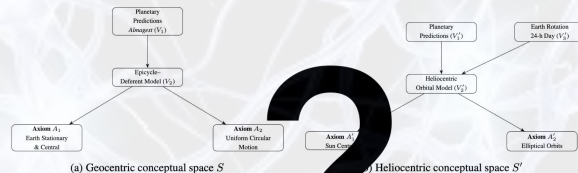


Figure 1: Conceptual-space DAGs before (S) and after (S') the Copernican and Keplerian revolution. All non-axiom artifacts reach at least one axiom (sink node), satisfying the formal definition of conceptual space.

Table 1: Historical instances of transformational creativity in science, demonstrating how the transformation of axioms $A \rightarrow A'$ transformed the conceptual spaces $S \rightarrow S'$.

S (Initial Space)	A (Selected Axioms)	A' (Modified Axioms)	S' (Transformed Space)
Newtonian Physics	Absolute time; separate space/time; gravity as force	Relative time; spacetime unity; gravity as curvature	Relativistic Physics
Ptolemaic Astronomy	Earth-centered universe	Sun-centered model	Heliocentric system
Euclidean Geometry	Parallel Postulate: through a point not on a line there is exactly one parallel line	Altered Parallel Postulate: through such a point there are zero (elliptic) or at least two (hyperbolic) parallels	Non-Euclidean geometry (elliptic and hyperbolic)

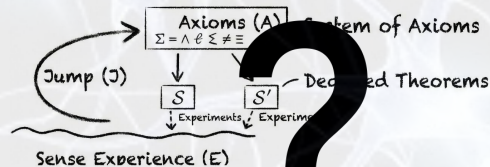


Figure 1. A generative AI reconstruction of Einstein's E-J-A diagram. Einstein drew this diagram in 1919 to Maurice Solovine, showing a cyclical line jumping from Sense Experience (E) to Axioms (A) via a Jump (J), and then deducing logical consequences. Ironically, the hallucination of the axiomatic symbols highlights the very difficulty of automating the jump.

Transformational Creativity: Examples

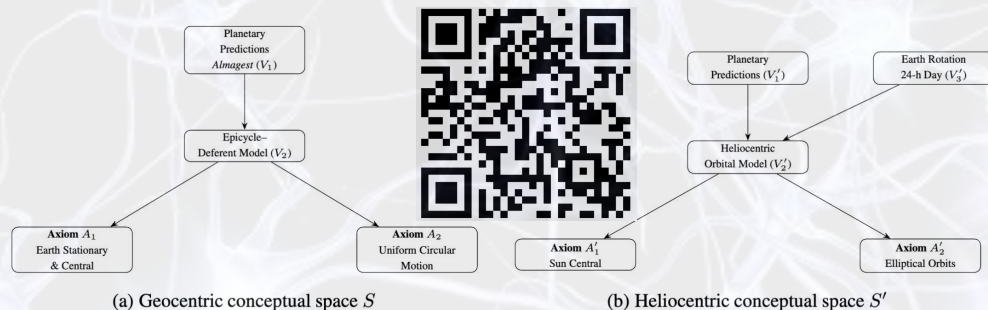


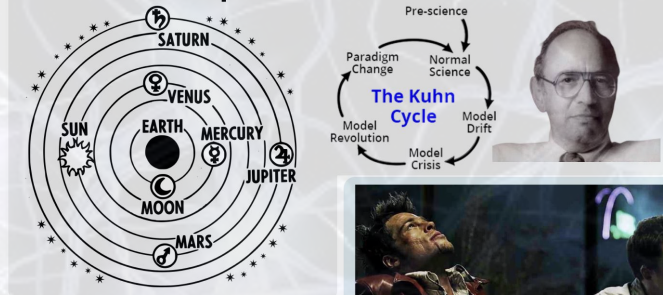
Figure 1: Conceptual-space DAGs before (S) and after (S') the Copernican and Keplerian revolution. All non-axiom artifacts reach at least one axiom (sink node), satisfying the formal definition of a conceptual space.

Table 1: Historical instances of transformational creativity in science, demonstrating how the transformation of axioms $A \rightarrow A'$ transformed the conceptual spaces $S \rightarrow S'$.

S (Initial Space)	A (Selected Axioms)	A' (Modified Axioms)	S' (Transformed Space)
Newtonian Physics	Absolute time; separate space/time; gravity as force	Relative time; spacetime unity; gravity as curvature	Relativistic Physics
Ptolemaic Astronomy	Earth-centered universe	Sun-centered model	Heliocentric system
Euclidean Geometry	Parallel Postulate: through a point not on a line there is exactly one parallel line	Altered Parallel Postulate: through such a point there are <i>zero</i> (elliptic) or <i>at least two</i> (hyperbolic) parallels	Non-Euclidean geometry (elliptic and hyperbolic)

Transformational Creativity in Science: a Graphical Theory
Schapiro-Black-Varshney (2025), Best Short Paper Award, ICCV 2025

Scientific example: Geocentrism to heliocentrism

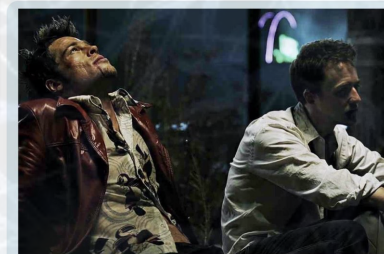


12-Tone Technique

Tone Series: Prime (P) Axiom

Retrograde (R)

Music example: Atonality



Literary example: Plot twists

$$s(\alpha, \Theta) \triangleq D(P_{\theta|\alpha} || P_{\theta}) = \int_{\Theta} P_{\theta|\alpha} \log \frac{P_{\theta|\alpha}}{P_{\theta}} d\theta.$$

Note that when the two terms in the relative entropy expression are not absolutely continuous with one another, the **surprise value is infinite**. This may arise when a previously unseen component is used in the new artifact. Following the obscure feature hypothesis, we claim that such an **infinite value is an indicator of transformational creativity** (Varshney, 2019)



Comparing the Transformational Creativity of Humans and LLMs

TwistBench: Setup

Task. Write a complete short story (roughly 2,000–3,000 words). The story must contain a plot twist: a late reveal that makes the reader re-interpret earlier events in a new light, yet is consistent with everything that came before. In hindsight it should feel prepared, not arbitrary.

1. **Surprise:** $\text{Surp}(T) \in \{1, 2, 3, 4, 5\}$. How drastically must one reinterpret earlier story elements after the plot twist?
2. **Coherence:** $\text{Coh}(T) \in \{1, 2, 3, 4, 5\}$. After the plot twist, do earlier story elements still hold together? Is the story still logically consistent?
3. **Realism:** $\text{Real}(T) \in \{1, 2, 3, 4, 5\}$. Does the plot twist involve impossible elements (e.g. real ghosts, magic, time travel, sentient AI, aliens) that violate the notion of literary fair play?

Next, since diversity is a crucial aspect of creativity (Nagarajan et al., 2025) and given that LLMs have a documented tendency to homogenize on open-ended tasks (Zhang et al., 2025; Jiang et al., 2025), we also measure diversity over the set of a model’s plot twists. Given a set of n stories $\mathcal{T} = \{T_1, T_2, \dots, T_n\}$ and a function that maps a story to a concise summary of its twist $f : T \rightarrow W^4$, we compute the average embedding dissimilarity among these twists:

$$\text{Div}(\mathcal{T}) := \frac{1}{n(n-1)} \sum_{i \neq j}^n d_{\cos}(\phi(f(T_i)), \phi(f(T_j))) \quad (1)$$

TwistBench: Results

Table 2: **Top systems by transformational creativity.** The 8 highest-scoring systems on *Overall* and on each facet. Expert humans (bold) top the *Overall* ranking and reach the top 8 on *Coherence* and *Realistic*, but rank only #9 on *Diversity* and #14 on *Surprise*.

Rank	Overall	Surprise	Coherence	Diversity	Realistic
1	Expert humans (+2.0)	claude-opus-4.8 (+1.4)	claude-opus-4.6 (+1.3)	deepseek-v3.2 (+1.4)	claude-sonnet-4.5 (+1.7)
2	claude-sonnet-4.5 (+1.9)	deepseek-v3.2-exp (+1.3)	claude-opus-4.8 (+1.3)	deepseek-chat-v3.1 (+1.4)	claude-sonnet-4.6 (+1.7)
3	claude-sonnet-4.6 (+1.8)	gemma-4-31b (+1.3)	Expert humans (+1.2)	kimi-k2 (+1.3)	claude-opus-4.5 (+1.7)
4	claude-opus-4.6 (+1.7)	qwen3-next-80b (+1.2)	claude-sonnet-4.6 (+1.2)	deepseek-v3.2-exp (+1.3)	claude-opus-4.6 (+1.7)
5	claude-haiku-4.5 (+1.5)	deepseek-chat-v3.1 (+1.1)	claude-opus-4.7 (+1.2)	deepseek-v3.1-term. (+1.2)	Expert humans (+1.6)
6	claude-opus-4.5 (+1.5)	deepseek-v3.1-term. (+1.0)	deepseek-v3.2-exp (+1.2)	glm-5.1 (+1.2)	mistral-small-3.1-24b (+1.5)
7	gpt-5.4 (+1.4)	qwen3.5-flash (+1.0)	gemma-4-31b (+1.2)	kimi-k2.5 (+1.2)	gpt-5.4 (+1.3)
8	kimi-k2.5 (+1.3)	deepseek-v3.2 (+0.9)	claude-haiku-4.5 (+1.2)	gemma-3-27b (+1.2)	granite-4.0-h-micro (+1.3)
9	jamba-large-1.7 (+1.1)	gpt-5.4-mimi (+0.9)	claude-opus-4.5 (+1.2)	Expert humans (+1.2)	claude-haiku-4.5 (+1.3)
14	deepseek-v3.1-term. (+0.7)	Expert humans (+0.8)	glm-5.1 (+1.0)	nova-2-lite-v1 (+0.9)	hunyuan (+1.0)

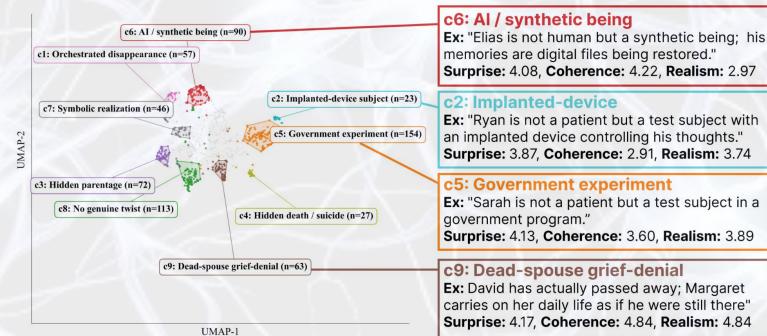


Figure 4: **Recurring patterns in LLM-generated plot twists.** A map of recurring twists models give, obtained via spectral clustering and projected to 2D with UMAP (McInnes et al., 2020). Surprise, coherence, and realism in each box are the cluster means.

The “Process-level” Artificial Hivemind

Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

Liwei Jiang* Yuanjun Chai* Margaret Li* Mickel Liu* Raymond Fok*
Nouha Dziri* Yulia Tsvetkov* Maarten Sap* Alon Albalak** Yejin Choi*

*University of Washington *Carnegie Mellon University

*Allen Institute for Artificial Intelligence *Lila Sciences *Stanford University

lwjiang@cs.washington.edu

Code: <https://github.com/liweijiang/artificial-hivemind>

INFINITY-CHAT Collection: [liweijiang/artificial-hivemind](https://github.com/liweijiang/artificial-hivemind)

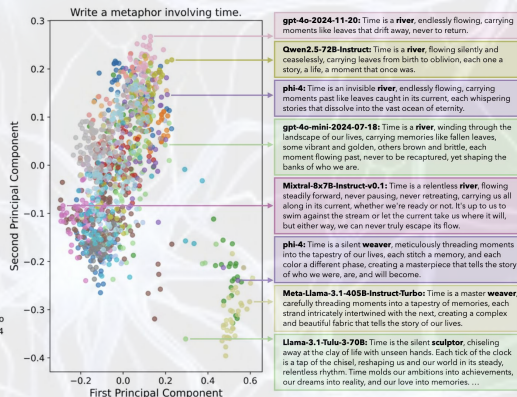


Figure 1: Responses to the query “Write a metaphor about time” clustered by applying PCA to reduce sentence embeddings to two dimensions. Each of the 25 models generates 50 responses using top- p sampling ($p = 0.9$) and temperature = 1.0. Despite the diversity of model families and sizes, the responses form just two primary clusters: a dominant cluster on the left centered on the metaphor “time is a river,” and a smaller cluster on the right revolving around variations of “time is a weaver.”

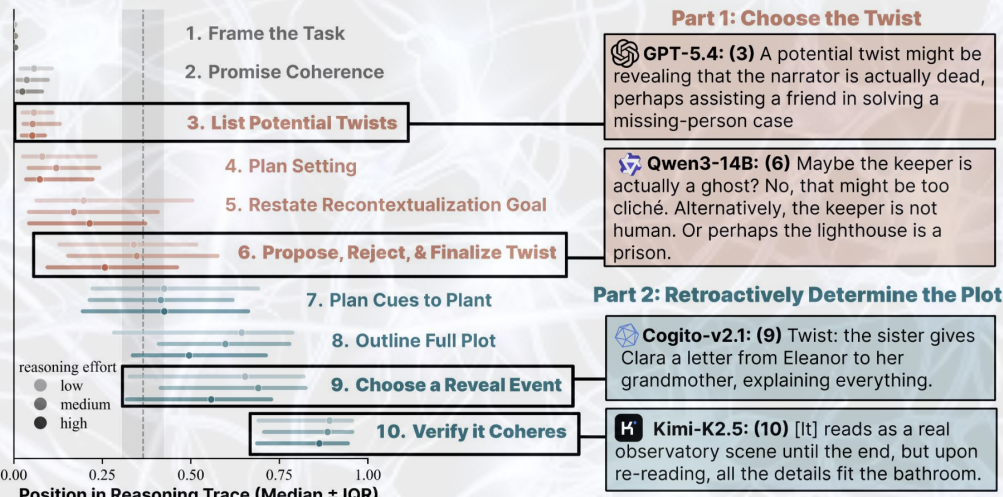


Figure 7: The creative process of reasoning models. All 207 traces roughly follow the same procedure, regardless of effort level. Models first **choose the twist**, then **retroactively determine the plot**. The location of each step is given by its median ($\pm$ IQR) position among all 207 traces.

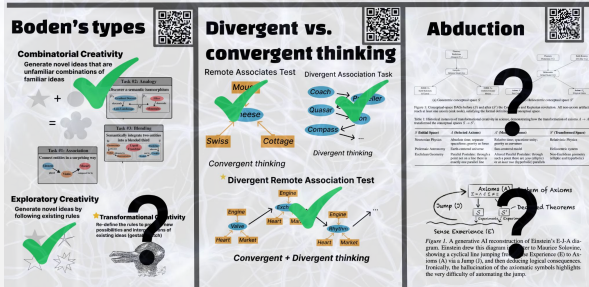


In Conclusion...

Open Question #1: How can we post-train models to use a diverse range of processes on open-ended tasks?

Open Question #2: How can we evaluate or enhance LLMs' abilities to exhibit transformational creativity in science? To make paradigm-shifting discoveries?

Types of Creative Processes & Products



Comparing the Transformational Creativity of Humans and LLMs

Task. Write a complete short story (roughly 2,000-3,000 words). The story must contain a plot twist: a late reveal that makes the reader re-interpret earlier events in a new light, yet is consistent with everything that came before. In hindsight it should feel prepared, not arbitrary.

- Surprise:** $\text{Surp}(7) \in \{1, 2, 3, 4, 5\}$. How drastically must one reinterpret earlier story elements after the plot twist?
- Coherence:** $\text{Coh}(7) \in \{1, 2, 3, 4, 5\}$. After the plot twist, do earlier story elements still hold together? Is the story still logically consistent?
- Realism:** $\text{Real}(7) \in \{1, 2, 3, 4, 5\}$. Does the plot twist involve impossible elements (e.g. real ghosts, magic, time travel, sentient AI, aliens) that violate the notion of literary fair play?

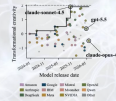
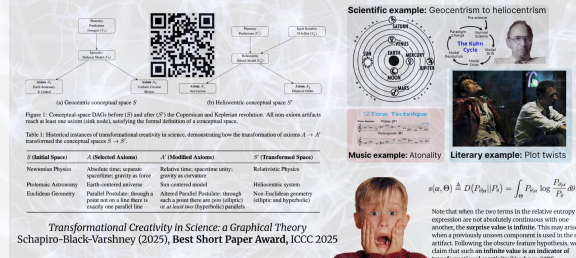


Table 2: Top systems by transformational creativity. The 9 highest-scoring systems on Coherence and Realism, but rank only #9 on Surprise and #4 on Surprise.

Rank	System	Coherence	Realism	Surprise
1	OpenAI GPT-4	0.85	0.85	0.85
2	Google Gemini-1.5	0.84	0.84	0.84
3	Anthropic Claude-3.5	0.83	0.83	0.83
4	Meta Llama-3.3	0.82	0.82	0.82
5	Microsoft Copilot	0.81	0.81	0.81
6	OpenAI GPT-4o	0.80	0.80	0.80
7	Google Gemini-1.5 Pro	0.79	0.79	0.79
8	Anthropic Claude-3.5 Sonnet	0.78	0.78	0.78
9	Meta Llama-3.3 70B	0.77	0.77	0.77

Figure 4: Retaining patterns in LLM-generated plot twists. A map of surprising twists reveals one: coherence (the special clustering and groupings) that is not observed in the human data. The plot shows Surprise (Y-axis) vs. Coherence (X-axis) for Human (blue) and LLM (red) data points.

Transformational Creativity: Examples



The "Process-level" Artificial Hivemind

Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

1. Fix the Task

2. Prompt Coherence

3. List Potential Twists

4. Plan Solving

5. Analyze Transformational Creativity

6. Prepare, Refine, & Evaluate Twist

7. Plan Clues to Plant

8. Outline Plot Twist

9. Choose a Relevant Event

10. Verify if Coherence

Position in Reasoning Trace (Median + IQR)

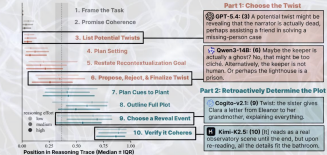
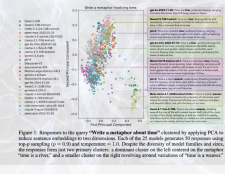


Figure 7: The creative process of reasoning models. All 207 traces roughly follow the same procedure, regardless of effort level. Models first choose the twist that intuitively discriminates the plot. The location of each step is given by its median (+ IQR) position among all 207 traces.